

The Coding Interview Patterns Playbook

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Course Introduction

Requirements

- ▶ Basic knowledge of any programming language - ideally Kotlin
- ▶ Optional: Understanding of the Gradle build system*
- ▶ **Preinstalled on computer:**
 - ▶ IntelliJ IDEA (Community Edition)**
 - ▶ Java 17***

* <http://udemy.com/course/mastering-gradle/>

* <https://www.jetbrains.com/idea/download/?section=mac>

** <https://adoptium.net/en-GB/temurin/releases/?version=17&os=mac&arch=aarch64&package=jre>

Welcome to the Course

- ▶ Why "Patterns" instead of "Problems"?
- ▶ Moving from memorization to intuition
- ▶ The 80/20 Rule: 19 patterns solve 80% of interviews

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How to Use This Course Effectively

- ▶ **Course Approach:** We will start by showing the brute-force solution, and then transition to the optimal solution.
- ▶ **In an Interview (Ideal Case):**
 1. **Identify:** Recognize pattern triggers in the problem text
 2. **Trace:** Dry-run the logic on a small input
 3. **Implement:** Code the solution without looking at hints

Interview Mindset: Thinking in Patterns

- ▶ Stop looking at the "Story" (e.g., "Alice has 3 apples...")
- ▶ Start looking at the **Data Flow**

Problem Trait	Pattern Strategy
Finding a sub-segment in a list	Sliding Window
Searching a sorted range	Binary Search
Shortest path in unweighted graph	BFS

Common Interview Mistakes

- ▶ **Mistake 1:** Starting to code before the logic is 100% clear
- ▶ **Mistake 2:** When stuck: avoiding brute force and trying to guess optimal solution
- ▶ **Mistake 3:** Forgetting to state Big O complexity
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Correction: Think out loud. Your interviewer is your collaborator, not your judge.

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Course Structure

- ▶ Pattern 1: Two Pointers Pattern
- ▶ Pattern 2: Sliding Window
- ▶ Pattern 3: Monotonic Stack
- ▶ Pattern 4: Prefix Sum & Cumulative
- ▶ Pattern 5: Cyclic Sort
- ▶ Pattern 6: Merge Intervals
- ▶ Pattern 7: Modified Binary Search
- ▶ Pattern 8: Fast & Slow Pointers
- ▶ Pattern 9: Tree Depth First Search (DFS)
- ▶ Pattern 10: Tree Breadth First Search (BFS)
- ▶ Pattern 11: Matrix Traversal (Islands)
- ▶ Pattern 12: Union-Find (Disjoint Set)
- ▶ Pattern 13: Top 'K' Elements (Heap)
- ▶ Pattern 14: Dynamic Programming
- ▶ Pattern 15: Backtracking & DFS
- ▶ Pattern 16: Greedy Algorithms
- ▶ Pattern 17: Data Streams & Online
- ▶ Pattern 18: Advanced Patterns
- ▶ Pattern 19: Miscellaneous Techniques

Fundamentals & Big O

The Absolute Essentials

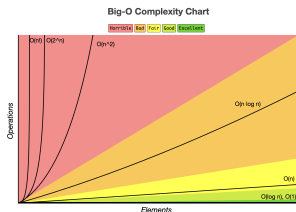
Before mastering coding patterns, you must master the fundamental building blocks.

- ▶ **Why?** Interviewers evaluate your ability to select the right tool for the job.
- ▶ **Key Areas:**
 - ▶ Analyzing efficiency (Big O Complexity).
 - ▶ Choosing the right data structures.
 - ▶ Writing idiomatic, clean code under pressure.

Big O Notation Demystified

How does the runtime or space requirement grow as input size N increases?

- ▶ **Time Complexity:** Count dominant operations.
 - ▶ $O(1)$ - Constant time.
 - ▶ $O(\log N)$ - Logarithmic (halving search space).
 - ▶ $O(N)$ - Linear (single pass).
 - ▶ $O(N \log N)$ - Linearithmic (sorting).
 - ▶ $O(N^2)$ - Quadratic (nested loops).
- ▶ **Space Complexity:** Measure auxiliary memory (call stack, maps, arrays).



biggocheatsheet.com

Growth Rates in Action ($N = 1,000,000$)

What do these complexities mean in practice for 1 million input elements?

Complexity	Operations ($N = 10^6$)	Approx. Time (1GHz CPU)
$O(1)$	1	≈ 1 ns (Instant)
$O(\log N)$	≈ 20	≈ 20 ns (Instant)
$O(N)$	1,000,000	≈ 1 ms (Instant)
$O(N \log N)$	$\approx 2 \times 10^7$	≈ 20 ms (Instant)
$O(N^2)$	10^{12}	≈ 16.7 minutes (Timeout)
$O(2^N)$	$2^{1,000,000}$	$\gg$ Age of the Universe (Impossible)

Data Structures Crash Course

Data Structure	Strengths	Common Use Case
Array / List	Fast index access ($O(1)$)	Ordered traversal, sorting
HashMap / Map	Key lookup ($O(1)$ average)	Frequency counting, cache
HashSet / Set	Duplicate removal, search ($O(1)$)	Visited trackers, unique items
Queue / Deque	FIFO ordering ($O(1)$ push/pop)	BFS, level order traversal

Top Kotlin Idioms at a Glance

Idiom / Construct	Primary Use Case	Codebase Example
<code>intArrayOf</code> / <code>arrayOf</code>	Matrix/test declarations	<code>intArrayOf(2, 7, 11, 15)</code>
<code>until</code> operator	Forward loop (0 to <i>size</i> - 1)	<code>for (i in 0 until nums.size)</code>
Safe Call (<code>?.</code>)	Null-safe node traversal	<code>cur1 = cur1?.next</code>
<code>kotlin.math</code> (<code>min</code> / <code>max</code>)	Range clamping	<code>kotlin.math.min(price, minPrice)</code>
<code>mutableListOf</code> / <code>listOf</code>	Building outputs/collections	<code>val res = mutableListOf<List<Int>>()</code>
<code>getOrDefault</code>	Map access with fallback	<code>adj.getOrDefault(v, hashMapOf())</code>
<code>ArrayDeque</code>	BFS queue / Stack	<code>val deque = ArrayDeque<IntArray>()</code>
<code>return if</code> (Expression)	Conditional returns	<code>return if (val1 >= 0) val1 else val2</code>

Pattern 1: Two Pointers Pattern

What is the Two Pointers Pattern?

- ▶ **The Core Idea:** Use two indices (pointers) to iterate through a data structure simultaneously.
- ▶ **Why it works:** It reduces $O(n^2)$ nested loops to a single $O(n)$ pass.

Pattern Mechanics: Opposite Ends

When searching for a target in a **Sorted** array:

1. **Left Pointer:** Starts at index 0.
2. **Right Pointer:** Starts at index $n - 1$.
3. **Decision:**
 - ▶ If $sum > target$: Move Right pointer left ($\leftarrow$) to decrease sum.
 - ▶ If $sum < target$: Move Left pointer right ($\rightarrow$) to increase sum.

Problem: Zero-Sum Triplets

Scenario: Given an integer array, find all unique triplets $[a, b, c]$ such that their sum equals exactly zero. The solution set must not contain duplicate triplets.

Examples:

- ▶ **Input:** $[-1, 0, 1, 2, -1, -4]$ → **Output:** $[-1, -1, 2], [-1, 0, 1]$
- ▶ **Input:** $[0, 1, 1]$ → **Output:** $[]$

Problem: Maximizing Container Area

Scenario: Given an array representing wall heights, find two walls that, together with the horizontal axis, form a container holding the maximum volume of water.

Examples:

- ▶ **Input:** [1, 8, 6, 2, 5, 4, 8, 3, 7] → **Output:** 49
Explanation: The walls at indices 1 and 8 have heights 8 and 7, width is 7. $\text{Area} = \min(8,7) * 7 = 49$.

Problem: Unique Elements In-Place

Scenario: Given a sorted array, remove duplicate elements in-place so that each unique element appears only once. Return the number of unique elements.

Examples:

▶ **Input:** [1, 1, 2] → **Output:** 2 (modified array: [1, 2, _])

Pattern 1: Two Pointers Pattern – Common Triggers

- ▶ The array is **Sorted**.
- ▶ You need to find a pair, a triplet, or a subarray.
- ▶ You need to "filter" or "rearrange" an array in-place.

Pattern 2: Sliding Window

What is the Sliding Window Pattern?

- ▶ **The Core Idea:** Maintain a contiguous window over a linear data structure (array/string) that dynamically expands or contracts.
- ▶ **Why it works:** Converts nested $O(n^2)$ loops searching subarrays into a single $O(n)$ pass by reusing computed information.

Dynamic vs. Fixed Window

- ▶ **Fixed Window:** Size K is constant. Move window by adding next element and removing the leftmost.
- ▶ **Dynamic Window:** Size changes.
 - ▶ Expand the window by moving 'right' pointer.
 - ▶ If constraint is violated, contract window by moving 'left' pointer until constraint is satisfied again.

Problem: Longest Unique Substring

Scenario: Given a string, find the length of the longest contiguous substring that contains only unique characters.

Examples:

- ▶ **Input:** 'abcabcbb' → **Output:** 3 ('abc')
- ▶ **Input:** 'bbbbbb' → **Output:** 1 ('b')

Problem: Anagram Substring Search

Scenario: Given two strings `s1` and `s2`, determine if `s2` contains a substring that is a permutation (anagram) of `s1`.

Examples:

- ▶ **Input:** `s1 = 'ab', s2 = 'eidbaooo'` → **Output:** `true`
- ▶ **Input:** `s1 = 'ab', s2 = 'eidboao'` → **Output:** `false`

Problem: Maximum Segment with Two Types

Scenario: Given an array of items, find the maximum length of a contiguous subarray that contains at most two distinct types of items.

Examples:

- ▶ **Input:** [1, 2, 1] → **Output:** 3
- ▶ **Input:** [0, 1, 2, 2] → **Output:** 3 ('[1, 2, 2]')

Problem: Smallest Window Containing Subsequence

Scenario: Given strings S and T , find the minimum length window substring of S which contains all characters of T (including duplicates). Return empty if no such window exists.

Examples:

▶ **Input:** $S = \text{'ADOBECODEBANC'}$, $T = \text{'ABC'}$ → **Output:**
 'BANC'

Pattern 2: Sliding Window – Common Triggers

- ▶ Subarray or substring problems.
- ▶ Finding shortest, longest, or target length matching constraint.
- ▶ Tracking unique elements/frequencies in a range.

Pattern 3: Monotonic Stack

What is a Monotonic Stack?

- ▶ **The Core Idea:** A stack that maintains elements in a strictly increasing or decreasing order.
- ▶ **How it works:** When a new element is processed, we pop all elements from the stack that violate the monotonic property before pushing the new element.

Monotonic Stack: Types

- ▶ **Monotonic Increasing Stack:** Stack elements are sorted in ascending order from bottom to top (e.g. '[1, 3, 5, 8]'). Useful to find the first smaller element to the left/right.
- ▶ **Monotonic Decreasing Stack:** Stack elements are sorted in descending order from bottom to top (e.g. '[9, 7, 4, 2]'). Useful to find the first larger element to the left/right.

Problem: Days to Warmer Temperature

Scenario: Given an array of daily temperatures, calculate the number of days you must wait until a warmer day occurs. If no warmer day is possible, output 0.

Examples:

▶ **Input:** [73, 74, 75, 71, 69, 72, 76, 73] → **Output:**
[1, 1, 4, 2, 1, 1, 0, 0]

Problem: Next Greater Element Search

Scenario: Given two arrays `nums1` and `nums2` where `nums1` is a subset of `nums2`, find the first element in `nums2` to the right of each `nums1` element that is strictly greater.

Examples:

- ▶ **Input:** `nums1 = [4,1,2]`, `nums2 = [1,3,4,2]` →
Output: `[-1, 3, -1]`

Problem: Volume of Trapped Water

Scenario: Given an elevation map of bar heights where the width of each bar is 1, calculate the total volume of water trapped between the bars after a heavy rain.

Examples:

▶ **Input:** [0, 1, 0, 2, 1, 0, 1, 3, 2, 1, 2, 1] →
Output: 6

Problem: Largest Rectangular Area in Histogram

Scenario: Given an array of integers representing heights of consecutive bars in a histogram, find the area of the largest rectangle that can fit inside the histogram boundary.

Examples:

▶ **Input:** [2, 1, 5, 6, 2, 3] → **Output:** 10

Pattern 3: Monotonic Stack – Common Triggers

- ▶ Finding the "next greater" or "previous smaller" element in an array.
- ▶ Computing areas bounded by heights (e.g. histograms, trapping water).
- ▶ Resolving queries in $O(N)$ total time where each index is pushed/popped at most once.

Pattern 4: Prefix Sum & Cumulative Patterns

What is the Prefix Sum Pattern?

- ▶ **The Core Idea:** Precompute cumulative sums of an array: 'prefix[i] = nums[0] + ... + nums[i]'.
- ▶ **Why it works:** Allows you to calculate the sum of any subarray 'nums[i...j]' in $O(1)$ time:

$$\text{Sum}(i, j) = \text{prefix}[j] - \text{prefix}[i - 1]$$

Combining Prefix Sum with HashMap

To find subarrays summing to K in $O(N)$ time:

- ▶ Maintain running cumulative sum 'currentSum'.
- ▶ At each step, we look for a previous prefix sum that equals 'currentSum - K '.
- ▶ Use a HashMap to store frequencies of prefix sums seen so far.
- ▶ **Key Modulo Trick:** For divisibility by K , store 'currentSum

Problem: Subarrays Summing to Target

Scenario: Given an array of integers and a target value k , return the total count of contiguous subarrays whose elements sum up exactly to k .

Examples:

▶ **Input:** `nums = [1, 1, 1]`, `k = 2` → **Output:** 2

Problem: Subarrays Sum Divisible by K

Scenario: Given an array of integers and a value k , return the number of non-empty contiguous subarrays that have a sum divisible by k .

Examples:

- ▶ **Input:** `nums = [4, 5, 0, -2, -3, 1]`, `k = 5` →
Output: 7

Pattern 4: Prefix Sum & Cumulative Patterns – Common Triggers

- ▶ Range sum query problems with static arrays.
- ▶ Finding subarrays that sum to a target value K .
- ▶ Subarrays with properties related to divisibility or modulo arithmetic.

Pattern 5: Cyclic Sort

What is the Cyclic Sort Pattern?

- ▶ **The Core Idea:** When an array contains numbers in a range from 1 to n (or 0 to n), we can sort it in $O(n)$ time and $O(1)$ auxiliary space.
- ▶ **How it works:** Iterate through the array. If the current number ' $\text{nums}[i]$ ' is not at its correct index (i.e. ' $\text{nums}[i] \neq i + 1$ '), swap it with the number at its correct index ' $\text{nums}[\text{nums}[i] - 1]$ '. Repeat this until the correct number is at index ' i ', then move to the next index.

Cyclic Sort Mechanics

- ▶ **Index Rule:** A value 'val' belongs at index 'val - 1' (or index 'val' if range is 0 to n).
- ▶ **Loop condition:** We only swap when the target index has a different value:

$$\text{nums}[i] \neq \text{nums}[\text{nums}[i] - 1]$$

- ▶ **Complexity analysis:** Although we have a nested loop or repeat swaps at the same index, each swap puts at least one number in its correct position. Thus, at most $N - 1$ swaps are performed, leading to $O(N)$ time.

Problem: Disappeared Numbers in Range

Scenario: Given an array of length n where each element is in the range $[1, n]$, find all integers in that range that do not appear in the array.

Examples:

► **Input:** $[4, 3, 2, 7, 8, 2, 3, 1]$ → **Output:** $[5, 6]$

Problem: First Missing Positive Integer

Scenario: Given an unsorted integer array, find the smallest positive integer that is not present in the array. Must run in $O(N)$ time and $O(1)$ auxiliary space.

Examples:

- ▶ **Input:** [1, 2, 0] → **Output:** 3
- ▶ **Input:** [3, 4, -1, 1] → **Output:** 2

Problem: Duplicate Elements in Range

Scenario: Given an array of integers of size n where $1 \leq \text{nums}[i] \leq n$ and some elements appear twice, find all duplicate elements.

Examples:

► **Input:** [4, 3, 2, 7, 8, 2, 3, 1] → **Output:** [2, 3]

Pattern 5: Cyclic Sort – Common Triggers

- ▶ Problems dealing with numbers in a continuous range.
- ▶ Finding duplicate or missing numbers in an array.
- ▶ Finding the first/smallest missing positive number.

Pattern 6: Merge Intervals

What is the Merge Intervals Pattern?

- ▶ **The Core Idea:** Process overlapping intervals by sorting them and merging adjacent ranges.
- ▶ **Why it works:** Once sorted by start time, overlapping intervals will be adjacent. We can merge them in a single linear pass.

Merging Logic

Given two sorted intervals A and B , they overlap if:

$$B.start \leq A.end$$

If they overlap, we merge them into a single interval C where:

$$C.start = A.start$$

$$C.end = \max(A.end, B.end)$$

Otherwise, they do not overlap, and we add A to our output list and set B as the new active interval.

Problem: Consolidating Overlapping Intervals

Scenario: Given an array of intervals [start, end], merge all overlapping intervals and return a list of non-overlapping intervals covering the same range.

Examples:

▶ **Input:** `[[1,3],[2,6],[8,10],[15,18]]` → **Output:**
`[[1,6],[8,10],[15,18]]`

Problem: Inserting into Non-Overlapping Intervals

Scenario: Given a set of non-overlapping intervals sorted by start time, insert a new interval and merge if it overlaps with existing intervals.

Examples:

▶ **Input:** `intervals = [[1,3],[6,9]]`, `newInterval = [2,5]` → **Output:** `[[1,5],[6,9]]`

Problem: Common Available Time Slots

Scenario: Given schedules of multiple users containing busy intervals, find the free time intervals common to all users.

Examples:

▶ **Input:** `[[[1,2],[5,6]], [[1,3]], [[4,10]]]` → **Output:**
`[[3,4]]`

Pattern 6: Merge Intervals – Common Triggers

- ▶ Finding overlapping ranges, free slots, or busy schedules.
- ▶ Inserting new intervals into already sorted ranges.
- ▶ Finding intersections of multiple lists of intervals.

Pattern 7: Modified Binary Search

What is Modified Binary Search?

- ▶ **The Core Idea:** An extension of standard binary search that applies to non-standard sorted structures or search spaces.
- ▶ **Why it works:** By dividing the search space in half in $O(1)$ time, we achieve a logarithmic $O(\log N)$ runtime.

Binary Search on Answer Space

Instead of searching indices in an array, we search value ranges (from 'minPossibleAnswer' to 'maxPossibleAnswer'):

- ▶ Find the midpoint 'mid' of the current range.
- ▶ Run a validation function 'isValid(mid)' to check if 'mid' is a possible solution (usually in $O(N)$ time).
- ▶ Based on the result:
 - ▶ If valid: Try to find a better solution in the lower half (if minimizing).
 - ▶ If invalid: Discard the lower half and search the upper half.

Problem: Search in Shifted Sorted Array

Scenario: Search for a target value in a sorted array that has been rotated (shifted circular-wise) at an unknown pivot index. Return its index, or -1 if not found.

Examples:

- ▶ **Input:** `nums = [4,5,6,7,0,1,2]`, `target = 0` →
Output: 4

Problem: Find Minimum in Shifted Sorted Array

Scenario: Given a sorted array that has been rotated (shifted circular-wise) at an unknown pivot index, find the minimum element in the array.

Examples:

► **Input:** [3, 4, 5, 1, 2] → **Output:** 1

Problem: Package Shipping Optimization

Scenario: A conveyor belt carries packages that must be shipped in order within D days. Find the minimum weight capacity of the ship that permits this.

Examples:

- ▶ **Input:** `weights = [1,2,3,4,5,6,7,8,9,10]`, `days = 5`
→ **Output:** 15

Problem: Fast Power Calculation

Scenario: Implement a function to calculate x raised to the power n , i.e., x^n , supporting negative integer powers.

Examples:

- ▶ **Input:** $x = 2.00$, $n = 10 \rightarrow$ **Output:** 1024.00
- ▶ **Input:** $x = 2.10$, $n = 3 \rightarrow$ **Output:** 9.261

Pattern 7: Modified Binary Search – Common Triggers

- ▶ Searching in a rotated sorted array.
- ▶ Finding boundaries, peaks, or transition points.
- ▶ **Binary Search on Answers:** When checking if a candidate answer is valid takes $O(N)$ time, and the answer is monotonic, we search the answer range.

Pattern 8: Fast & Slow Pointers

What is the Fast & Slow Pointers Pattern?

- ▶ **The Core Idea:** Use two pointers traversing a linear data structure (linked list/array) at different speeds.
- ▶ **Why it works:** If there is a cycle, the fast pointer (moving 2 steps at a time) will eventually meet the slow pointer (moving 1 step at a time) in $O(N)$ time and $O(1)$ space.

Cycle Detection Mechanics

- ▶ **Initialize:** 'slow = head', 'fast = head'.
- ▶ **Advance:**
 - ▶ 'slow = slow.next'
 - ▶ 'fast = fast.next.next'
- ▶ **Check:** If 'slow == fast', a cycle is detected! If 'fast' or 'fast.next' is null, there is no cycle.
- ▶ **Cycle Start (Floyd's algorithm):** Once met, reset 'slow' to 'head'. Move both pointers 1 step at a time. The point where they meet again is the start of the cycle.

Problem: Cycle Detection in Linked Nodes

Scenario: Determine if a linked node sequence contains a cycle where a node's next pointer points back to a previous node.

Examples:

- ▶ **Input:** head = [3,2,0,-4], cycle at index 1 →
Output: true

Problem: Center Node of Link Sequence

Scenario: Given a singly linked list, find and return the middle node. If there are two middle nodes, return the second middle node.

Examples:

- ▶ **Input:** [1, 2, 3, 4, 5] → **Output:** Node(3)
- ▶ **Input:** [1, 2, 3, 4, 5, 6] → **Output:** Node(4)

Problem: Find Repeated Number in Constant Space

Scenario: Given an array of size $n+1$ where each integer is in range $[1, n]$, find the repeated number using only constant extra space and without modifying the array.

Examples:

► **Input:** $[1, 3, 4, 2, 2]$ → **Output:** 2

Pattern 8: Fast & Slow Pointers – Common Triggers

- ▶ Cycle detection in a linked list.
- ▶ Finding the starting node of a cycle.
- ▶ Finding the middle node of a list (when the fast pointer reaches the end, the slow pointer is in the middle).
- ▶ Identifying duplicates in structures where values can be treated as indices.

Pattern 9: Tree Depth First Search (DFS)

Tree Depth First Search (DFS)

- ▶ **The Core Idea:** Explore as deep as possible along each branch of a tree before backtracking.
- ▶ **Implementation:** Typically implemented using recursion (which uses the call stack implicitly) or an explicit Stack.

DFS Traversal Types

Depending on when the current node is processed relative to its children:

- ▶ **Pre-order (Root, Left, Right):** Processing parent before children. Useful for copying trees or serialization.
- ▶ **In-order (Left, Root, Right):** Processing left child, then parent, then right child. Yields sorted keys in a Binary Search Tree (BST).
- ▶ **Post-order (Left, Right, Root):** Processing children before parent. Useful for bottom-up computation (e.g., node height, subproblem accumulation).

Problem: Root-to-Leaf Target Sum

Scenario: Given a binary tree and a targetSum, return true if the tree has a root-to-leaf path such that adding up all the values along the path equals targetSum.

Examples:

- ▶ **Input:** root =
[5,4,8,11,null,13,4,7,2,null,null,null,1], target
= 22 → **Output:** true

Problem: Any-Path Sum Count

Scenario: Given a binary tree, find the number of paths that sum to a given target value. The path does not need to start or end at the root or leaf, but must go downwards.

Examples:

► **Input:** root = [10,5,-3,3,2,null,11,3,-2,null,1],
target = 8 → **Output:** 3

Problem: Validate Binary Search Tree Structure

Scenario: Determine if a given binary tree is a valid Binary Search Tree (BST), where left children are smaller and right children are larger.

Examples:

- ▶ **Input:** [2, 1, 3] → **Output:** true
- ▶ **Input:** [5, 1, 4, null, null, 3, 6] → **Output:** false

Problem: Lowest Common Ancestor

Scenario: Find the lowest common ancestor node of two given nodes in a binary tree.

Examples:

▶ **Input:** root = [3,5,1,6,2,0,8], p = 5, q = 1 →

Output: 3

Problem: Maximum Path Sum in Tree

Scenario: Find the maximum path sum in a binary tree, where a path is any sequence of nodes from some starting node to any node in the tree along parent-child connections.

Examples:

▶ **Input:** [-10, 9, 20, null, null, 15, 7] → **Output:**
42

Pattern 9: Tree Depth First Search (DFS) – Common Triggers

- ▶ Finding paths that sum to a target value.
- ▶ Computing properties of nodes recursively (height, balance).
- ▶ Validating BST properties.
- ▶ Exploring combinations/subtrees.

Pattern 10: Tree Breadth First Search (BFS)

Tree Breadth First Search (BFS)

- ▶ **The Core Idea:** Traverse a tree level-by-level, processing all nodes at depth d before any node at depth $d + 1$.
- ▶ **Implementation:** Typically implemented iteratively using a Queue (FIFO).

Level Size Strategy

To process nodes level by level rather than as a continuous queue stream:

- ▶ At the start of each level iteration, count the queue size: `'val levelSize = queue.size'`.
- ▶ Run a nested loop exactly `'levelSize'` times.
- ▶ Inside the loop:
 - ▶ Pop a node from the queue.
 - ▶ Enqueue its children (left, then right).
- ▶ Any logic scoped to a level (e.g., calculating level average or reversing order) can be done cleanly.

Problem: Level-by-Level Tree Traversal

Scenario: Given a binary tree, return the level order traversal of its nodes' values (from left to right, level by level).

Examples:

▶ **Input:** [3, 9, 20, null, null, 15, 7] → **Output:**
[[3], [9, 20], [15, 7]]

Problem: Zigzag Level Order Traversal

Scenario: Given a binary tree, return the zigzag level order traversal of its nodes' values (left-to-right, then right-to-left for the next level, and alternate).

Examples:

▶ **Input:** [3, 9, 20, null, null, 15, 7] → **Output:**
[[3], [20, 9], [15, 7]]

Problem: Connecting Right Neighbors in Tree

Scenario: Given a perfect binary tree, populate each next pointer to point to its next right node. If there is no next right node, set it to NULL.

Examples:

► **Input:** [1, 2, 3, 4, 5, 6, 7] → **Output:** [1, #, 2, 3, #, 4, 5, 6, 7, #]

Problem: Word Transformation Sequence

Scenario: Given two words (`beginWord`, `endWord`) and a dictionary, find the length of the shortest transformation sequence from `beginWord` to `endWord`, changing one letter at a time.

Examples:

▶ **Input:** `begin = 'hit', end = 'cog', wordList = ['hot', 'dot', 'dog', 'lot', 'log', 'cog']` → **Output:** 5

Pattern 10: Tree Breadth First Search (BFS) – Common Triggers

- ▶ Level order traversals or printing.
- ▶ Finding the minimum depth of a tree.
- ▶ Tracking "level" boundaries (e.g. connecting next pointers).
- ▶ Finding the shortest path in unweighted graphs or state transitions.

Pattern 11: Matrix Traversal (The "Islands" Pattern)

What is the Matrix Traversal Pattern?

- ▶ **The Core Idea:** Treat a 2D grid (matrix) as an implicit graph where each cell (r, c) is a node, and its adjacent neighbors (up, down, left, right) are connected by edges.
- ▶ **Traversal:** Use Depth First Search (DFS) or Breadth First Search (BFS) to explore connected regions.

Matrix Traversal Mechanics

- ▶ **Boundary Check:** Always ensure row 'r' and column 'c' are valid:

$$0 \leq r < R \quad \text{and} \quad 0 \leq c < C$$

- ▶ **Directions Array:** Use a utility array to iterate over neighbors cleanly:

$$\text{dirs} = [(-1, 0), (1, 0), (0, -1), (0, 1)]$$

- ▶ **Visited Tracking:** Mark grid cells as visited (e.g., set "1" to "0" in-place if allowed, or use a 'visited' boolean matrix) to prevent infinite loops.

Problem: Counting Isolated Grid Clusters

Scenario: Given a 2D grid of '1's (land) and '0's (water), count the number of isolated land clusters (islands) surrounded by water cardinally.

Examples:

► **Input:** grid =
[['1','1','0'],['1','1','0'],['0','0','0']] →
Output: 1

Problem: Largest Connected Grid Cluster

Scenario: Find the maximum area of a connected grid cluster (island) in a 2D grid, where area is the number of cardinally connected land cells.

Examples:

► **Input:** `grid = [[0, 1], [1, 1]]` → **Output:** 3

Problem: Spreading Contagion in Grid

Scenario: In a 2D grid containing fresh items, contaminated items, and empty spaces, find the minimum time elapsed until all fresh items are contaminated. Spreads cardinally. Return -1 if impossible.

Examples:

► **Input:** grid = `[[2,1,1],[1,1,0],[0,1,1]]` → **Output:**
4

Problem: Water Flow Boundaries

Scenario: Given a 2D matrix of heights, find all coordinates from which water can flow cardinally to both the left/top ocean (Pacific) and the right/bottom ocean (Atlantic).

Examples:

▶ **Input:** heights = `[[1,2],[2,1]]` → **Output:**
`[[0,0],[0,1],[1,0],[1,1]]`

Pattern 11: Matrix Traversal (The "Islands" Pattern) – Common Triggers

- ▶ Counting distinct blocks (e.g. islands, provinces).
- ▶ Simulating cell expansion over time (e.g. rotting oranges, fire spreading).
- ▶ Finding paths or reachable regions from boundaries.

Pattern 12: Union-Find (Disjoint Set)

What is the Union-Find Pattern?

- ▶ **The Core Idea:** A data structure that tracks partition of a set into disjoint subsets.
- ▶ **Operations:**
 - ▶ **Find:** Identify which subset an element belongs to (returns root parent).
 - ▶ **Union:** Join two disjoint subsets into a single subset.

Optimizations: Path Compression & Union by Rank

Without optimization, a tree can become a line ($O(N)$ lookup).

- ▶ **Path Compression:** During 'find(x)', make every node in the search path point directly to the root root. This flattens the tree structure:

$$\text{parent}[x] = \text{find}(\text{parent}[x])$$

- ▶ **Union by Rank/Size:** Always attach the smaller tree under the root of the larger tree.
- ▶ **Complexity:** Combined, they achieve nearly constant time operations: $O(\alpha(N))$ where α is the Inverse Ackermann function.

Problem: Connected Components in Network

Scenario: Given an adjacency matrix representing relationships (e.g. friendship), find the total number of connected groups/components in the network.

Examples:

- ▶ **Input:** `isConnected = [[1,1,0],[1,1,0],[0,0,1]]` →
Output: 2

Problem: Locating Cycle-Creating Edge

Scenario: Given an undirected graph with n nodes and n edges containing exactly one cycle, find the edge that can be removed to turn the graph into a tree.

Examples:

► **Input:** $[[1,2], [1,3], [2,3]] \rightarrow$ **Output:** $[2,3]$

Problem: Merging Overlapping Identifiers

Scenario: Given a list of accounts with names and emails, merge accounts that share common emails, returning a sorted list of merged emails.

Examples:

► **Input:** `[['John', 'a@m.com', 'b@m.com'],
['John', 'c@m.com', 'a@m.com']]` → **Output:**
`[['John', 'a@m.com', 'b@m.com', 'c@m.com']]`

Pattern 12: Union-Find (Disjoint Set) – Common Triggers

- ▶ Dynamic connectivity (are nodes X and Y in the same group?).
- ▶ Detecting cycles in undirected graphs.
- ▶ Merging accounts, groups, or equivalence sets.

Pattern 13: Top 'K' Elements (Heap / Priority Queue)

What is the Top 'K' Elements Pattern?

- ▶ **The Core Idea:** Find the K largest, smallest, or most frequent elements in a large collection.
- ▶ **Why it works:** Instead of sorting the entire array in $O(N \log N)$ time, we can maintain a Heap/Priority Queue of size K .
- ▶ **Heaps selection rule:**
 - ▶ To find the **K largest** elements: Use a **Min-Heap**. When size exceeds K , pop the smallest element. The heap will retain the K largest values.
 - ▶ To find the **K smallest** elements: Use a **Max-Heap**. When size exceeds K , pop the largest.
- ▶ **Complexity:** $O(N \log K)$ time and $O(K)$ extra space.

Problem: K Most Frequent Elements

Scenario: Given an integer array and an integer k , return the k most frequent elements in the array. You may return the answer in any order.

Examples:

▶ **Input:** `nums = [1,1,1,2,2,3]`, `k = 2` → **Output:** `[1, 2]`

Problem: K Nearest Coordinates to Origin

Scenario: Given an array of points on the X-Y plane and an integer k , return the k closest points to the origin $(0, 0)$ using Euclidean distance.

Examples:

▶ **Input:** `points = [[1,3],[-2,2]]`, `k = 1` → **Output:** `[[-2,2]]`

Problem: CPU Task Scheduler

Scenario: Given a list of CPU tasks represented by characters and a cooldown period n , find the minimum CPU intervals required to complete all tasks.

Examples:

- ▶ **Input:** `tasks = ['A','A','A','B','B','B'], n = 2` →
Output: 8

Problem: Rearrange No-Adjacent-Duplicate Characters

Scenario: Given a string, rearrange its characters so that any two adjacent characters are not the same. Return empty string if impossible.

Examples:

- ▶ **Input:** 'aab' → **Output:** 'aba'
- ▶ **Input:** 'aaab' → **Output:** ''

Pattern 14: Dynamic Programming

What is Dynamic Programming (DP)?

- ▶ **The Core Idea:** Solve complex problems by breaking them down into simpler, overlapping subproblems and caching their results to avoid redundant calculations.
- ▶ **Key Properties:**
 - ▶ **Overlapping Subproblems:** Solving the same subproblems repeatedly.
 - ▶ **Optimal Substructure:** The optimal solution of the problem can be constructed from optimal solutions of its subproblems.
- ▶ **Approaches:**
 - ▶ **Top-Down (Memoization):** Recursive, start from target, solve subproblems and save values in a cache.
 - ▶ **Bottom-Up (Tabulation):** Iterative, start from base cases, fill a table (array/matrix) up to target.

Dynamic Programming Sub-patterns

Most interview DP problems fall into standard subclasses:

- ▶ **Fibonacci Numbers:** $dp[i] = dp[i-1] + dp[i-2]$ (Climbing Stairs).
- ▶ **0/1 Knapsack:** Items can be chosen at most once. Decide to include or exclude an item: $dp[i][w] = \max(\text{exclude}, \text{include})$.
- ▶ **Unbounded Knapsack / Coin Change:** Items can be chosen multiple times.
- ▶ **Longest Common Subsequence (LCS) / Edit Distance:** String matching.
- ▶ **Longest Increasing Subsequence (LIS):** Decisions based on previous elements.

Problem: Distinct Ways to Climb Stairs

Scenario: You are climbing a staircase that takes n steps to reach the top. Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?

Examples:

- ▶ **Input:** 2 → **Output:** 2
- ▶ **Input:** 3 → **Output:** 3

Concept: 0/1 Knapsack

Problem: Minimum Coins for Change

Scenario: Given coins of different denominations and a total amount, write a function to compute the fewest number of coins that you need to make up that amount.

Examples:

▶ **Input:** `coins = [1,2,5]`, `amount = 11` → **Output:** 3

Problem: Combinations count for Change

Scenario: Given coins of different denominations and a total amount, return the number of combinations that make up that amount.

Examples:

► **Input:** coins = [1,2,5], amount = 5 → **Output:** 4

Problem: Longest Strictly Increasing Subsequence

Scenario: Given an integer array `nums`, return the length of the longest strictly increasing subsequence.

Examples:

▶ **Input:** `[10,9,2,5,3,7,101,18]` → **Output:** 4

Problem: String Segmentation using Dictionary

Scenario: Given a string and a dictionary of words, return true if the string can be segmented into a space-separated sequence of dictionary words.

Examples:

▶ **Input:** `s = 'leetcode', wordDict = ['leet','code']`
→ **Output:** `true`

Pattern 15: Backtracking & DFS

What is the Backtracking Pattern?

- ▶ **The Core Idea:** A systematic search algorithm that builds candidate solutions incrementally, and abandons a candidate ("backtracks") as soon as it determines the candidate cannot lead to a valid solution.
- ▶ **How it works (The Three Steps):**
 1. **Choose:** Add an element to the current path.
 2. **Explore:** Recursively call the search function with the new state.
 3. **Unchoose (Backtrack):** Remove the element from the current path to return to the previous state.

Problem: Well-Formed Parentheses Combinations

Scenario: Given n pairs of parentheses, write a function to generate all combinations of well-formed parentheses.

Examples:

► **Input:** 3 → **Output:**

`['((()))', '(()())', '(())()', '()(())', '()()()']`

Problem: Combination Sum to Target

Scenario: Given an array of distinct integers candidates and a target integer target, return a list of all unique combinations of candidates where the chosen numbers sum to target.

Examples:

- ▶ **Input:** candidates = [2,3,6,7], target = 7 →
Output: [[2,2,3],[7]]

Problem: Palindrome Substring Partitions

Scenario: Given a string, partition s such that every substring of the partition is a palindrome. Return all possible palindrome partitioning of s .

Examples:

▶ **Input:** 'aab' → **Output:** [['a', 'a', 'b'], ['aa', 'b']]

Problem: Word Grid Search

Scenario: Given an $m \times n$ grid of characters and a string word, return true if word exists in the grid. The word can be constructed from letters of sequentially adjacent cells.

Examples:

► **Input:** board =
[['A', 'B', 'C', 'E'], ['S', 'F', 'C', 'S']], word =
'ABCCED' → **Output:** true

Problem: N-Queens Placement

Scenario: Place n queens on an $n \times n$ chessboard such that no two queens attack each other. Return all distinct solutions.

Examples:

► **Input:** 4 → **Output:** `[['.Q..', '...Q', 'Q...',
'..Q.'], ['..Q.', 'Q...', '...Q', '.Q..']]`

Pattern 15: Backtracking & DFS – Common Triggers

- ▶ Generating all permutations, combinations, or subsets.
- ▶ Solving constraint satisfaction puzzles (e.g. N-Queens, Sudoku).
- ▶ Exploring word combinations on a grid.

Pattern 16: Greedy Algorithms

What are Greedy Algorithms?

- ▶ **The Core Idea:** An algorithmic paradigm that builds up a solution piece by piece, always choosing the next piece that offers the most obvious and immediate local benefit ("greedy" choice).
- ▶ **Why it works:** For certain problems, making locally optimal decisions leads to a globally optimal solution.
- ▶ **Tradeoffs:**
 - ▶ Highly efficient (usually $O(N)$ or $O(N \log N)$).
 - ▶ Difficult to mathematically prove correctness.
 - ▶ If greedy fails, you must fall back to Dynamic Programming (which checks all combinations).

Problem: Circular Route Fuel Stations

Scenario: Given gas and cost arrays for circular route fuel stations, find the starting station index from which you can travel around the circuit once without running out of gas.

Examples:

- ▶ **Input:** gas = [1,2,3,4,5], cost = [3,4,5,1,2] →
Output: 3

Problem: Min Candy Distribution

Scenario: There are n children standing in a line. Each child has a rating. Distribute candies such that each child has at least 1, and children with higher ratings than neighbors get more.

Examples:

▶ **Input:** $[1, 0, 2]$ → **Output:** 5

Problem: Line Reconstruction by Height Attributes

Scenario: Reconstruct a queue where each person is represented by $[h, k]$ where h is their height and k is the number of people in front of them who are taller or equal.

Examples:

► **Input:** $[[7,0], [4,4], [7,1], [5,0], [6,1], [5,2]] \rightarrow$
Output: $[[5,0], [7,0], [5,2], [6,1], [4,4], [7,1]]$

Pattern 16: Greedy Algorithms – Common Triggers

- ▶ Scheduling, interval selection, minimizations.
- ▶ Gas stations, container loading, coin distributions (canonical cases).

Pattern 17: Data Streams & Online Algorithms

Data Streams & Online Algorithms

- ▶ **The Core Idea:** Process data that arrives continuously over time, rather than having the entire dataset available in memory upfront.
- ▶ **Constraint:** Algorithms must run in real-time with limited space ($O(1)$ or $O(K)$ where K is the query window size) and time complexity per element.
- ▶ **Key Techniques:**
 - ▶ **Two Heaps:** Balancing a Max-Heap and Min-Heap to find median in $O(1)$ time.
 - ▶ **Reservoir Sampling:** Selecting a random sample from a stream of unknown size.
 - ▶ **Double Linked List + HashMap:** Designing fast cache policies (like LRU/LFU cache) with $O(1)$ insertion, lookup, and deletion.

Problem: Running Median from Data Stream

Scenario: Design a data structure that supports adding numbers from a data stream and finding the median of the numbers added so far.

Examples:

- ▶ **Input:** `addNum(1)`, `addNum(2)`, `findMedian()` →
Output: 1.5

Problem: Random Pick Index from Duplicates

Scenario: Given an integer array with duplicate elements, randomly pick an index of a given target number with equal probability.

Examples:

▶ **Input:** `nums = [1,2,3,3,3]`, `pick(3)` → **Output:** 2, 3, or 4

Problem: Moving Average in Sliding Window

Scenario: Given a stream of integers and a window size, calculate the moving average of all integers in the sliding window.

Examples:

▶ **Input:** size = 3, next(1), next(10), next(3), next(5) → **Output:** 1.0, 5.5, 4.66, 6.0

Problem: LRU Cache Design

Scenario: Design a data structure that follows the constraints of a Least Recently Used (LRU) cache with get and put operations in $O(1)$ time.

Examples:

► **Input:** capacity = 2, put(1,1), put(2,2), get(1), put(3,3) → **Output:** get(1)→1, get(2)→-1

Pattern 18: Advanced Patterns

Advanced Algorithmic Patterns

For Senior and Staff level roles, interviewers often present advanced structures:

- ▶ **Segment Tree:** A binary tree used for storing intervals or segments. It allows querying a range (sum, min, max) and modifying elements in $O(\log N)$ time.
- ▶ **Topological Sort:** Ordering vertices in a directed acyclic graph (DAG) such that for every directed edge uv , vertex u comes before v . (Kahn's BFS or DFS).
- ▶ **Sweep Line:** A geometry paradigm that solves range/interval problems by sorting boundary events (start/end) and sweeping a imaginary line across them.
- ▶ **Trie (Prefix Tree):** A search tree used to store associative keys (typically strings) efficiently. Excellent for autocompletion, spelling checkers, prefix matching.

Problem: Dynamic Array Range Sum Query

Scenario: Given an array, implement a structure to calculate range sum queries [left, right] and update elements at index in logarithmic time.

Examples:

- ▶ **Input:** `nums = [1,3,5]`, `sumRange(0,2)`, `update(1,2)`, `sumRange(0,2)` → **Output:** 9, then 8

Problem: Course Dependency Resolution

Scenario: Determine if you can finish all courses given the total number of courses and prerequisite dependencies.

Examples:

- ▶ **Input:** numCourses = 2, prerequisites = [[1,0]] → **Output:** true
- ▶ **Input:** numCourses = 2, prerequisites = [[1,0],[0,1]] → **Output:** false

Problem: Outline of Bounded Buildings

Scenario: Given building locations and heights, return the key points that outline the collective skyline contour.

Examples:

▶ **Input:** `[[2,9,10],[3,7,15]]` → **Output:**
`[[2,10],[3,15],[7,0]]`

Problem: Prefix Tree Implementation

Scenario: Implement a trie (prefix tree) with insert, search, and startsWith methods.

Examples:

▶ **Input:** insert('apple'), search('apple'), startsWith('app') → **Output:** true, true

Pattern 19: Miscellaneous Techniques

Miscellaneous Interview Techniques

Some interview questions don't fit into single patterns, but require a combination of general skills:

- ▶ **In-place Grid Simulation:** Modifying a matrix without using extra memory by using rows/columns headers as state markers (e.g. Set Matrix Zeroes).
- ▶ **Math & Coordinate Transformations:** Mirroring, swapping, or transposing grids (e.g. Rotate Image).
- ▶ **Stack Evaluation:** Parsing arithmetic strings or expressions sequentially (e.g. Evaluate Reverse Polish Notation).
- ▶ **Bitmasking or Row/Column hashing:** Validating board boundaries quickly (e.g. Valid Sudoku).

Problem: Spread Zeroes in 2D Array

Scenario: Given an $m \times n$ integer matrix, if an element is 0, set its entire row and column to 0 in-place.

Examples:

► **Input:** $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ → **Output:**
 $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

Problem: Validate Sudoku Board State

Scenario: Determine if a 9×9 Sudoku board configuration is valid, checking rows, columns, and 3×3 sub-grids.

Examples:

▶ **Input:** board = ... → **Output:** true/false

Problem: Rotate 2D Matrix In-Place

Scenario: Given an $n \times n$ 2D matrix representing an image, rotate the image by 90 degrees clockwise in-place.

Examples:

► **Input:** $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ → **Output:**
 $\begin{bmatrix} 7 & 4 & 1 \\ 8 & 5 & 2 \\ 9 & 6 & 3 \end{bmatrix}$

Problem: Postfix Expression Evaluation

Scenario: Evaluate the value of an arithmetic expression in Reverse Polish Notation (RPN), supporting standard operators.

Examples:

▶ **Input:** ['2', '1', '+', '3', '*'] → **Output:** 9

Section 22 — Summary & The Modern Interview Landscape

Systematic Problem-Solving Template

How to tackle any unknown problem in an interview:

1. **Clarify:** Ask questions about constraints (n , value limits, sorted status, memory limits).
2. **Example:** Dry run a custom, non-trivial test case.
3. **Brute Force:** State the obvious solution ($O(N^2)$ or $O(2^N)$), explaining its bottleneck.
4. **Optimize:** Apply pattern matching (e.g. "Do I need a window?", "Can a map help?").
5. **Code:** Write clean, modular code, stating edge cases.
6. **Dry Run:** Trace the code line-by-line with the example.

The Engineering Manager's Lens

Interviews are a signal of how you will perform as a team member:

- ▶ ****Tradeoffs:**** Be vocal about time vs. space complexity tradeoffs.
- ▶ ****Production-Ready Code:**** Favor clean modular helper methods, input validation, and proper typing over dense one-liners.
- ▶ ****Collaboration:**** Treat the interviewer as a teammate. Don't go silent for minutes; explain your thought flow continuously.

Redefining the SDLC in the AI Era

How AI coding agents (like Gemini, Copilot) change the baseline expectation:

- ▶ ****Syntax is Cheap:**** Being a human compiler or memorizing syntax is no longer the primary value.
- ▶ ****Logic & System Design is King:**** Focus on high-level reasoning, system architecture, identifying constraints, and verifying correctness.
- ▶ ****Coding speed vs. System security:**** Writing code quickly is easy; ensuring it is robust, secure, and fits into the existing framework is the real challenge.

Recap & Roadmap Beyond the Course

- ▶ You have studied 20+ crucial algorithmic patterns!
- ▶ ****Next Steps:****
 - ▶ Practice spaced repetition (solve a problem, review it after 3 days, then 7 days).
 - ▶ Conduct mock whiteboard interviews with peers.
 - ▶ Build intuition over syntax.

Congratulations on completing the course!